

Data Bible: Bolt Datathon

1. Overview

The Vancouver City FC has been facing a decline in revenue across three pillars: Stadium operations, Fanbase engagement, and Merchandise sales, as well as issues engaging international audiences. We aim to optimize these sources of revenue to modernize Vancouver City FC's business model.

- **Project:** Vancouver City FC Revenue Insight
- **Source:** Bolt Case Package
- **Date Range:** Dec. 2023 - Dec. 2024

2. Data Sources

Source Name	Description	Format
fanbase_clean.csv	Member profiles and demographics	CSV
cleaned_merch.csv	Purchase history and item categories	CSV
clean_stadium.csv	Stadium operations incl. revenue sources	CSV
merch_fanbase_merged.csv	Merged file of merch and fanbase info (Outer, on Member ID)	CSV

3. Data Dictionary

Fanbase Engagement (fanbase_clean.csv) 70000 records

Column Name	Type	Description
Member_ID	integer	Unique membership identifier (ID)
Customer_Age_Group	string (Ordinal)	Age bracket
Customer_Region	string (Categorical)	Country or broader region the member is from
Seasonal_Pass	boolean	Whether customer holds a seasonal pass
Customer_Age_Group		

	Value
	'<18'
	'18-25'
	'26-40'
	'41-60'
	'60+'

Customer_Region

Value
'Canada'
'USA'
'Japan'
'India'
'China'
'South Korea'
'UK'

Stadium Operations(clean_stadium.csv) 144 records

Column Name	Type	Description
Month	integer (Ordinal)	The month within the 2024 Calendar Year
Revenue	integer	Revenue amount generated by assigned source during that month
Source	string (Categorical)	Specific source of revenue or cost

Month

Code	Value
1	January
2	February
3	March
4	April
5	May
6	June

Code	Value
7	July
8	August
9	September
10	October
11	November
12	December

Source

Value
'Upper Bowl'
'Lower Bowl'
'Season'
'Premium'
'Food'
'Staff'
'Concert'

Merchandise Purchase (`cleaned_merch.csv`) 67877 records

Column Name	Type	Description
Product_ID	Integer (Categorical)	Unique product identifier
Barcode	Integer	The barcode (e.g., UPC/EAN) associated with the exact SKU sold to an individual customer; this is the number scanned at checkout.
Item_Category	Categorical	Type of merchandise
Item_Name	Categorical	Specific item name
Size	Categorical	Item size
Unit_Price	Integer (Ordinal)	Price per unit (CAD)
Customer_Age_Group	Categorical	Estimated age bracket of customer, may differ from purchaser age
Customer_Region	Categorical	Domestic vs international
Promotion	Boolean	Indicates whether a specific promotional campaign led to the purchase. If the customer

Column Name	Type	Description
		engaged with a tracked promotion (e.g., clicked an ad) and then purchased, this is TRUE; otherwise, it is FALSE.
Channel	Categorical	Sales channel
Selling_Date	Date	Date and time of transaction
Arrival_Date	Date	For online orders, the date and time the product was delivered to the customer. This field is left blank (or NA) for in-store purchases

Item_Category

Cap

Hoodie

Jersey

Mug

Poster

Scarf

Youth Hoodie

Youth Jersey

Product_ID	Item_Name
10000001	Classic Cap
10000002	Retro Cap
10000003	Training Hoodie
10000004	Home Jersey
10000005	Away Jersey
10000006	Retro Jersey
10000007	Club Mug
10000008	Logo Mug
10000009	Player Poster
10000010	Team Poster
10000011	Home Scarf
10000012	Away Scarf
10000013	Youth Team Hoodie
10000014	Youth Training Hoodie
10000015	Youth Away Jersey

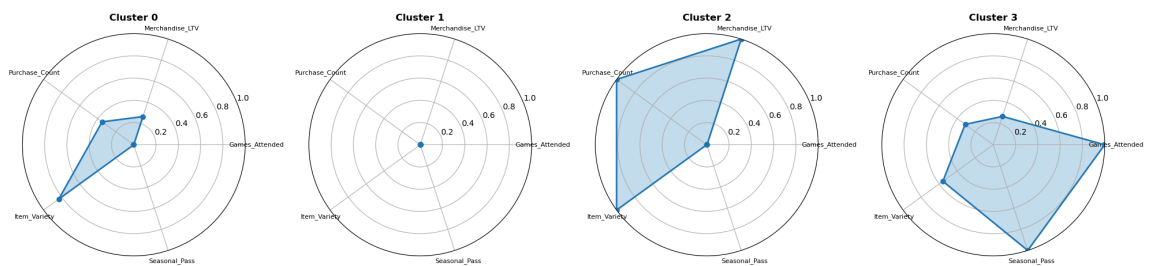
Product_ID	Item_Name
10000016	Youth Retro Jersey

Size
L
M
S
XL
XXL
YL
YS
YM
NaN

Customer_Age_Group
<18
18–25
26–40
41–60
60+

Channel
Online
Team Store

4. Cluster Profiles



Merch Enthusiasts

Metric	Value
Size	33,009 fans
Avg LTV	\$365.87
Avg Games	4.5
Avg Merch	\$123.13
Season Pass %	0.0%

Disengaged Fans

Metric	Value
Size	23,791 fans
Avg LTV	\$244.97
Avg Games	4.5
Avg Merch	\$0.00
Season Pass %	0.0%

Superfans/VIPs

Metric	Value
Size	8,451 fans
Avg LTV	\$706.89
Avg Games	4.5
Avg Merch	\$461.81
Season Pass %	0.0%

Loyal Attendees

Metric	Value
Size	4,749 fans
Avg LTV	\$1,342.59
Avg Games	22.5
Avg Merch	\$123.83
Season Pass %	100.0%
